

ELEC5305

Acoustics, Speech and Signal Processing (Project)

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1. Introduction

With the widespread adoption of human-computer interaction technology, speech recognition has become an indispensable core module in devices such as smartphones and smart speakers [1]. However, most high-performance speech recognition systems rely on large-scale labeled corpora and complex deep neural network structures, making them difficult to deploy directly on embedded platforms with limited computing and storage resources [2]. Therefore, achieving stable and reliable speech recognition under the constraint of "small dataset + lightweight model" is of great significance for both current industrial applications and scientific research [3].

The Free Spoken Digit Dataset (FSDD) is a very small, publicly available dataset containing digits 0–9 recorded by multiple speakers. However, performing speech recognition on such a small dataset faces at least two fundamental difficulties. First, the pronunciation differences between different speakers are significant, easily leading to underfitting or overfitting of individual speakers. Second, noise interference is unavoidable in real-world environments. Noise alters the time-frequency structure of speech, making traditional feature extraction methods less robust and significantly reducing recognition performance at low signal-to-noise ratios [4].

To address the aforementioned issues, this study compares the performance differences between MFCC+SVM and Log-Mel+CNN in the FSDD digit recognition task, focusing on the two core dimensions of "feature selection + model structure." To simulate background noise in real-world devices, this project adds additive white noise with different signal-to-noise ratios (SNR) to the original clean speech, constructing various noise scenarios and systematically examining the performance changes of the two schemes under noisy conditions.

The research questions of this project can be summarized as follows: On a small-scale corpus like the Free Spoken Digit Dataset, how do the traditional MFCC+SVM and the lightweight CNN based on Log-Mel spectrum perform in terms of recognition accuracy, noise robustness, and inference latency? Which scheme is more suitable as a lightweight digital speech recognition front-end for embedded scenarios?

The main contributions of this study include:

1. Analyzing the structural information loss caused by MFCC during dimensionality compression at the feature level, and empirically demonstrating its impact on SVM classification performance;
2. Training a lightweight CNN on FSDD to verify the matching advantage of Log-Mel as a two-dimensional time-frequency feature with convolutional structures, demonstrating that CNN can better utilize local patterns of time and frequency;
3. Comparing the performance of MFCC+SVM and Log-Mel+CNN under various SNR scenarios, and providing design suggestions for small datasets and embedded deployments based on recognition rate and latency.

2. Literature Review

MFCC is a classic speech feature that approximates the perceptual characteristics of the human ear through a Mel filter bank, and then further reduces the spectrum to a lower dimension through logarithmic compression and DCT [5]. Although this transformation can remove redundant information and improve the classification efficiency of shallow models, it also inevitably loses important time-frequency structural features. In addition, MFCC is a "static feature", that is, each frame is treated as an independent vector, and its temporal dependence is weakened, making SVM unable to exploit the dynamic changes of speech [6]. Therefore, when noise causes deformation or energy drift in some key frequency bands, the stability of MFCC will be significantly reduced, and SVM cannot automatically correct feature shifts from the context, ultimately leading to a sudden performance drop. From the feature level to the classifier level, MFCC+SVM is sensitive to structural noise [7].

Unlike the cepstral structure of MFCC, the Log-Mel spectrogram completely retains the original distribution of the speech signal in both time and frequency dimensions, making it more like a "continuously evolving acoustic image." This type of two-dimensional structure is very suitable for the convolution mechanism of CNN. The convolution kernel can automatically extract important features such as edges, textures, energy peaks, and frequency band changes from local time-frequency regions, effectively capturing the formant trajectory and phoneme morphology of speech. In addition, deep networks can automatically filter and enhance features in noise areas, giving the model stronger generalization capabilities [8].

3. Methodology

3.1 Dataset

This study uses Free Spoken Digit Dataset (FSDD) as the experimental data set. The data set is obtained from GitHub, as shown in Figure 3.1. The dataset consists of ten digits 0–9 recorded by multiple speakers. Each recording has a sampling rate of 8kHz and a duration of approximately 0.8–1.0 seconds. All audio is in single-channel WAV format. FSDD is characterized by clear sound quality and clean background, so it is suitable for studying the performance changes of digital speech recognition systems "before and after adding noise".

In this experiment, the data was divided into training set and test set at 80%:20%, and the test set was kept fixed in all noise experiments to ensure fair comparison of different models under the same conditions. In addition, to avoid additional data bias, all audio is normalized, truncated, or padded before being input into the model to ensure consistent length and adaptability to subsequent feature extraction methods.

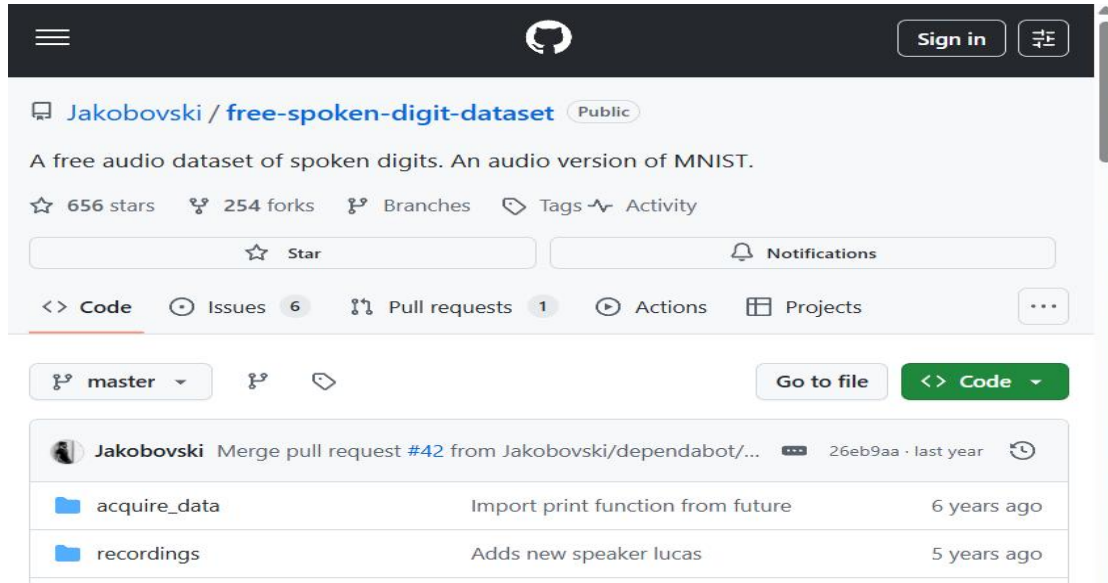


Fig3.1 dataset source from github

3.2 Feature Extraction

This study builds traditional models and deep learning models respectively, so two different types of feature extraction methods are used: MFCC features commonly used in traditional speech recognition, and Log-Mel spectral features that are standard in deep learning models.

The MFCC feature extracts the short-time spectral envelope of speech by simulating the Mel frequency scale perceived by the human ear, which is the core feature of traditional speech recognition systems. In this study, after performing pre-emphasis processing on each piece of audio, a 25ms frame length and a 10ms frame shift were used for frame windowing, and then a fast Fourier transform (FFT) was performed on each frame to obtain the power spectrum. The energy of each frequency band was calculated and logarithmically calculated through the Mel filter bank (a total of 26 filters), and finally the 13-dimensional static MFCC coefficients were obtained through the discrete cosine transform (DCT). In order to enhance the ability to express dynamic features, this study adds first-order difference (Δ) and second-order difference ($\Delta\Delta$) at the same time, making the final feature dimension 39 dimensions. All MFCC features are flattened into fixed-length vectors and fed into the SVM classifier for completing the ten-class task.

Log-Mel is a standard input feature for convolutional neural network (CNN) speech recognition systems. Different from MFCC, Log-Mel retains the complete time-frequency structure. This study first performs short-time Fourier transform (STFT) on the speech signal to obtain the time-frequency distribution matrix, then uses 40 Mel filters to calculate the energy of each frequency band and takes the logarithm to generate the Log-Mel spectrogram. Compared with MFCC, Log-Mel features not only have frequency domain smoothness, but also retain local time-frequency patterns, so they are very suitable for extracting local structures such as formant distribution and syllable fluctuations of speech through the convolution kernel of CNN. The final Log-Mel spectrogram obtained in this study is a two-dimensional matrix of approximately 40×80 , and channel dimensions are added as needed before inputting the CNN to comply with the input requirements of the convolutional neural network.

3.3 Model Design

This study selects two types of typical representative models: traditional machine learning methods (MFCC+SVM) and small convolutional neural networks based on deep learning (Log-Mel+CNN) to analyze the differences between the two types of methods in structure, training mechanism, and robustness to noise.

Support vector machine (SVM) is a small sample classification model suitable for low-dimensional features. In the training phase, SVM receives the flattened 39-dimensional MFCC features and maps them to high-dimensional space through the kernel function to complete the linear separability of digital categories. This study adjusted parameters through a simple grid search.

Convolutional neural network (CNN) is the current mainstream lightweight speech recognition model. The CNN structure constructed in this study includes two layers of convolutional layers (both using 3×3 convolution kernels), two layers of max pooling layers, one fully connected layer and the final softmax classification layer. The model uses Adam optimizer, cross-entropy loss function, the number of training rounds is set to 20–30, and the validation set loss is observed during the training process to avoid overfitting.

3.4 Noise Injection Experiment

In order to systematically evaluate the robustness of the model in the actual environment, this study adds additive white noise (AWGN) with different signal-to-noise ratios (SNR) to the test set audio, including six scenarios of 0, 5, 10, 15, 20 dB and clean. Noise injection is implemented using MATLAB's `awgn()` function, so that the signal energy and noise energy of each test audio segment meet the specified SNR value. All noise is only added to the test set, while the training set is kept completely clean to analyze the generalization ability of the model "without seeing noise". By testing the recognition performance of SVM and CNN under six SNRs, we can observe the performance change trends of the two models under weak noise, strong noise, and even very low SNR, and analyze the fundamental difference in the sensitivity of the models to noise.

3.5 Evaluation Metrics

In order to comprehensively evaluate the model performance, this study uses accuracy, confusion matrix and inference latency as test indicators. The accuracy is used to measure the overall accuracy of classification, the confusion matrix is used to analyze the model's misjudgment structure between different digital categories, and the inference delay is used to measure the real-time performance of the CNN in actual deployment. This study measures the average inference time of CNN through multiple repeated tests and records its delay changes under different SNR conditions to analyze whether noise will cause additional overhead in the model calculation process.

4. Results

4.1 Baseline SVM (MFCC) result analysis

In the baseline experiments of this study, we performed a small-scale grid hyperparameter search using an RBF kernel support vector machine (SVM) based on the eigenvectors composed of MFCCs and their differential coefficients. As shown in Figure 4.1, the cross-validation (CV) accuracy of the 9 groups of combinations shows an obvious trend with parameter changes: when kernelScale=0.5, the results of the 9 groups all stay at a low level of 0.288 to 0.760. When kernelScale=1, the performance is improved, and some combinations reach 0.664~0.760. When kernelScale=2 and C=2, the model achieved the highest verification accuracy of 94.2% in this set of experiments. This shows that the separability of MFCC features between different numerical classes is limited. A larger kernelScale can construct a smoother and more covering high-dimensional decision boundary, while a higher C can effectively reduce the misclassification tolerance caused by soft intervals. The combination of the two improves the classification ability within the training distribution.

```
ks=0.50, C=0.50 -> CV acc=0.288
ks=0.50, C=1.00 -> CV acc=0.288
ks=0.50, C=2.00 -> CV acc=0.218
ks=1.00, C=0.50 -> CV acc=0.760
ks=1.00, C=1.00 -> CV acc=0.758
ks=1.00, C=2.00 -> CV acc=0.664
ks=2.00, C=0.50 -> CV acc=0.929
ks=2.00, C=1.00 -> CV acc=0.936
ks=2.00, C=2.00 -> CV acc=0.942
Best CV acc = 0.942
```

Fig4.1 SVM Hyperparameter Search

Although the model under the optimal parameters achieved an extremely high accuracy of 99.95% on the training set and 86.67% on the validation set, the actual performance on the test set was only 57.33%, showing significant overfitting. As shown in Figure 4.2, the number 1 still maintains a high recognition rate, but numbers such as 3, 5, 7 are seriously confused. This reflecting the model's inability to maintain consistent MFCC distribution discriminative capabilities among unseen speakers.

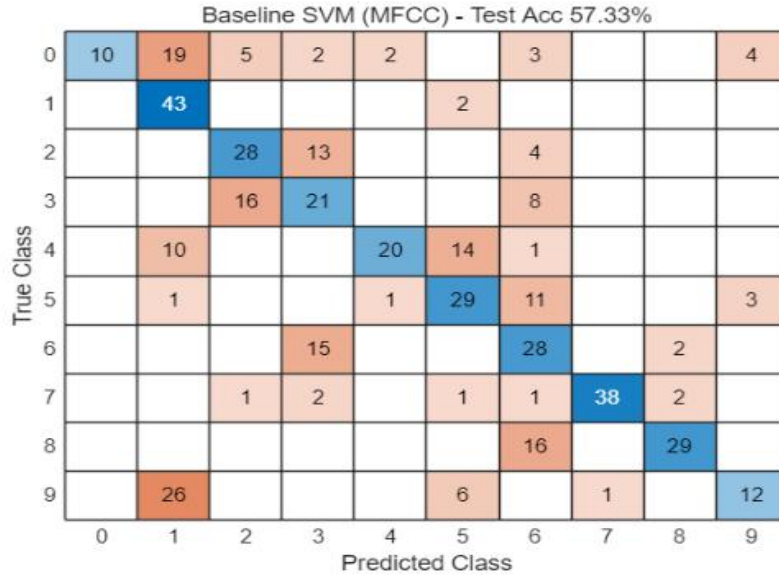


Fig4.2 Confusion Matrix of Baseline SVM Classifier

Since SVM uses static MFCC features instead of time series features, it cannot capture important dynamic information that changes over time in digital speech, especially for numbers with long pronunciation structures and obvious changes (such as "2", "4", and "6").

These phenomena jointly indicate that although the combination of MFCC + SVM performs well in the training set, it lacks the ability to generalize to real recording scenarios and is difficult to meet the cross-speaker consistency and robustness requirements of digital speech recognition tasks.

4.2 Analysis of experimental results of CNN (Log-Mel input)

In this study, the convolutional neural network (CNN) model based on log-mel spectrogram converged after about 20 rounds of training. Its training process and performance are shown in Figure 4.3. The mini-batch accuracy of the model at the initial stage of training is only about 4.69%, and the verification accuracy is about 11.56%, accompanied by high training and verification losses, indicating that the network is extremely weak in recognizing complex acoustic patterns when it is not fully learned. However, as the number of training rounds increases, the accuracy continues to increase rapidly. By the 20th round, the model training accuracy was stable at 95.31%, the verification accuracy was about 91.11%, indicating that the network had reached a good fitting state and there were no obvious signs of overfitting. The learning curve chart also shows that the training and verification accuracy rates are close to each other with small fluctuations, further verifying the stability of the model.

Epoch	Step	Elapsed Time (hh:mm:ss)	Train Accuracy	Val Accuracy	Train Loss	Val Loss	Base LR
1	1	0:00:00	4.69%	11.56%	4.0072	3.134	0.001
1	30	0:00:01	21.88%	22.22%	2.24	2.1894	0.001
2	50	0:00:01	25.00%	—	2.0792	—	0.001
2	60	0:00:01	23.44%	32.67%	1.961	1.8855	0.001
3	90	0:00:02	54.69%	56.89%	1.3577	1.3578	0.001
4	100	0:00:02	40.62%	—	1.4441	—	0.001
4	120	0:00:03	73.44%	65.56%	0.9516	0.9516	0.001
5	150	0:00:03	70.31%	79.78%	0.8115	0.645	0.001
6	180	0:00:03	73.44%	82.89%	0.7822	0.5009	0.001
7	200	0:00:03	81.25%	—	0.5734	—	0.001
7	210	0:00:04	92.19%	87.11%	0.4006	0.4317	0.001
8	240	0:00:04	87.50%	88.00%	0.3956	0.3798	0.001
8	250	0:00:04	78.12%	—	0.5361	—	0.001
9	270	0:00:04	85.94%	88.22%	0.4662	0.3672	0.001
10	300	0:00:05	92.19%	88.00%	0.2808	0.3642	0.001
11	330	0:00:06	93.75%	89.56%	0.2161	0.3355	0.001
11	350	0:00:06	90.62%	—	0.2528	—	0.001
12	360	0:00:06	90.62%	90.44%	0.1965	0.2844	0.001
13	390	0:00:07	92.19%	88.00%	0.24	0.3943	0.001
13	400	0:00:07	93.75%	—	0.1596	—	0.001
14	420	0:00:07	89.06%	90.22%	0.2834	0.2369	0.001
15	450	0:00:07	90.62%	90.89%	0.2109	0.2623	0.001
15	480	0:00:08	100.00%	91.56%	0.0467	0.2706	0.001
16	500	0:00:08	98.44%	—	0.0683	—	0.001
16	510	0:00:08	92.19%	91.56%	0.1985	0.2856	0.001
17	540	0:00:09	96.88%	93.11%	0.116	0.2378	0.001
18	550	0:00:09	93.75%	—	0.1256	—	0.001
18	570	0:00:09	100.00%	90.89%	0.0539	0.2534	0.001
19	600	0:00:10	98.44%	90.89%	0.0606	0.2495	0.001
20	630	0:00:10	95.31%	90.44%	0.0837	0.2522	0.001
20	640	0:00:11	96.88%	91.11%	0.0814	0.2529	0.001

Fig4.3 CNN training log

During the testing phase of the model, CNN achieved a classification accuracy of 74.00% on the unseen test set. As shown in Figure 4.4. It is significantly better than the aforementioned baseline SVM model using MFCC features (57.33%), indicating that the log-mel spectrogram as a two-dimensional convolution input can more effectively capture the time-frequency structural characteristics of speech data and is a feature form with higher expressive power. Its confusion matrix also reflects the stronger recognition ability of CNN: on most categories, such as numbers 1, 2, 4, 7, 8, etc., the model achieves a high diagonal hit rate, showing that the model can accurately capture its characteristic patterns on these categories; however, there is still a certain degree of confusion in some categories, such as 0, 3, 6, 9, etc.

In addition to accuracy, this experiment also evaluates the operating efficiency of the model. By running 20 repeated tests, the average delay of the CNN model's prediction of a single sample is only 0.0282 ms. The inference speed is extremely fast and fully meets the needs of real-time application scenarios. If deployed to embedded devices or online identification systems, its latency performance will have obvious advantages.

Average CNN latency per sample over 20 runs: 0.0282 ms

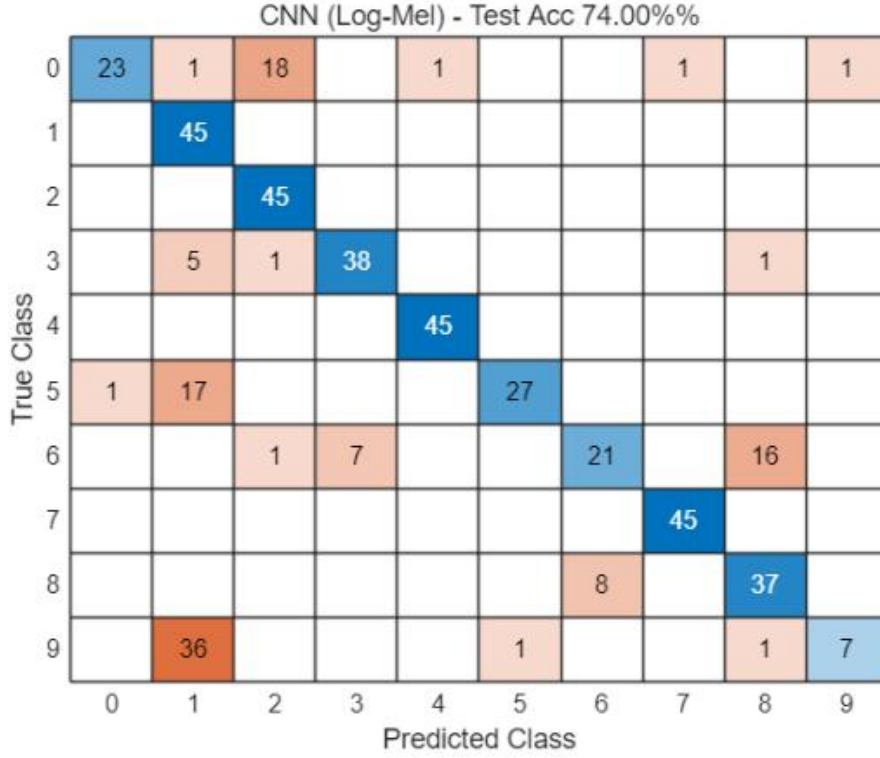


Fig4.4 Confusion Matrix of CNN Classifier

Taken together, the CNN (log-mel) model in this experiment performs well in terms of accuracy, stability, and inference speed, and is significantly improved compared to the traditional SVM + MFCC baseline model. The training log shows a regular and clear convergence process, the loss function has a stable downward trend, and the final test accuracy reaches 74%, proving that the convolutional network combined with log-mel features is an effective speech classification solution, and its low-latency characteristics ensure that the model has good practical deployment potential.

4.3 Description of noise robustness test results

In the noise robustness experiment, this study compared and evaluated the anti-noise capabilities of the MFCC-based SVM model and the log-mel spectrogram-based CNN model under the conditions of SNR = 0, 5, 10, 15, and 20 dB. The experimental results are shown in Figure 4.5. Overall, the accuracy of the SVM model remains at about 10% under all noise levels, which is basically equivalent to random guessing, indicating that it is extremely sensitive to noise and difficult to perform effective audio classification in high-noise environments.

```

Noise test: SNR = 0 dB
  SVM (MFCC) accuracy 0 dB = 10.00%%
  CNN (log-mel) accuracy 0 dB = 16.67%%, latency = 0.030 ms/sample
Noise test: SNR = 5 dB
  SVM (MFCC) accuracy 5 dB = 10.00%%
  CNN (log-mel) accuracy 5 dB = 20.44%%, latency = 0.028 ms/sample
Noise test: SNR = 10 dB
  SVM (MFCC) accuracy 10 dB = 10.00%%
  CNN (log-mel) accuracy 10 dB = 31.56%%, latency = 0.030 ms/sample
Noise test: SNR = 15 dB
  SVM (MFCC) accuracy 15 dB = 10.00%%
  CNN (log-mel) accuracy 15 dB = 45.78%%, latency = 0.029 ms/sample
Noise test: SNR = 20 dB
  SVM (MFCC) accuracy 20 dB = 10.00%%
  CNN (log-mel) accuracy 20 dB = 61.78%%, latency = 0.025 ms/sample

```

Fig4.5 Comparison of SVM and CNN under different SNR

In contrast, although the accuracy of the CNN model drops significantly under low SNR conditions, it still maintains significantly higher performance than SVM: reaching 16.67% at 0 dB, increasing to about 20% at 5 dB, and exceeding 31% at 10 dB, showing a trend of gradual improvement as SNR increases. When the SNR increases to 15 dB and 20 dB, the CNN accuracy increases to about 46% and 61% respectively, showing good noise robustness. Especially in high signal-to-noise ratio environments, CNN can utilize the time-frequency features of log-mel spectrogram to maintain stable classification performance, which is significantly better than SVM that only relies on low-dimensional MFCC features.

As shown in Figure 4.6, this study further plotted the classification accuracy under different SNR conditions into a line chart to visually compare the performance difference between SVM (MFCC features) and CNN (log-mel features) in a noisy environment.

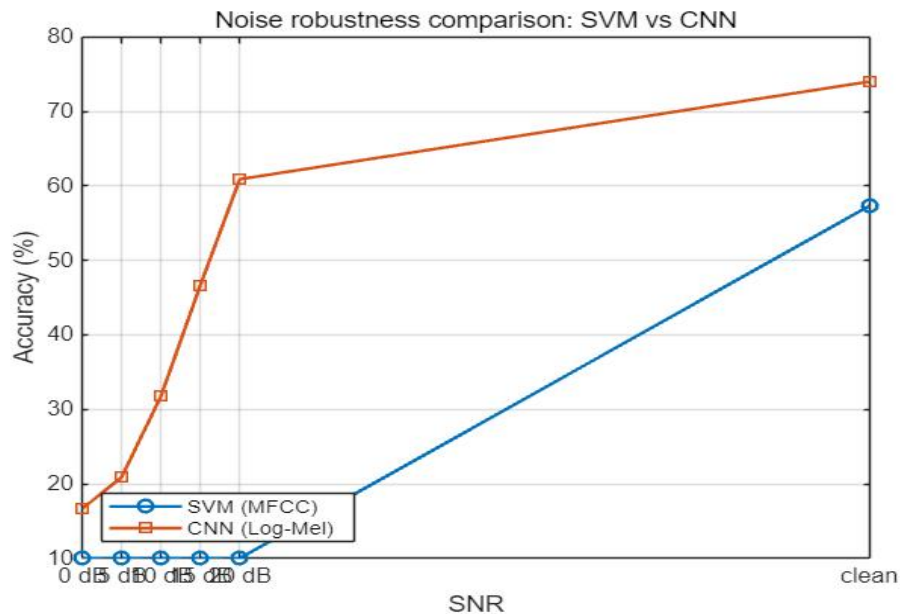


Fig4.6 Noise robustness curves: SVM vs CNN

Figure 4.7 shows the variation trend of the average inference latency of the CNN model with SNR under different noise levels (SNR = 0–20 dB).

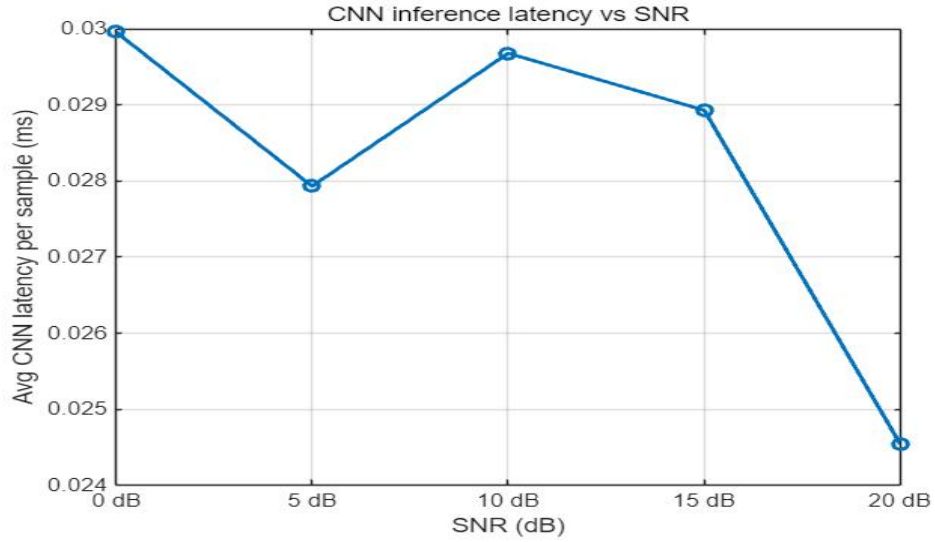


Fig4.7 CNN inference latency as a function of SNR

Figure 4.8 shows the changing trend of recognition accuracy of SVM (based on MFCC features) and CNN (based on log-mel spectrogram) under different noise levels (SNR = 0–20 dB).

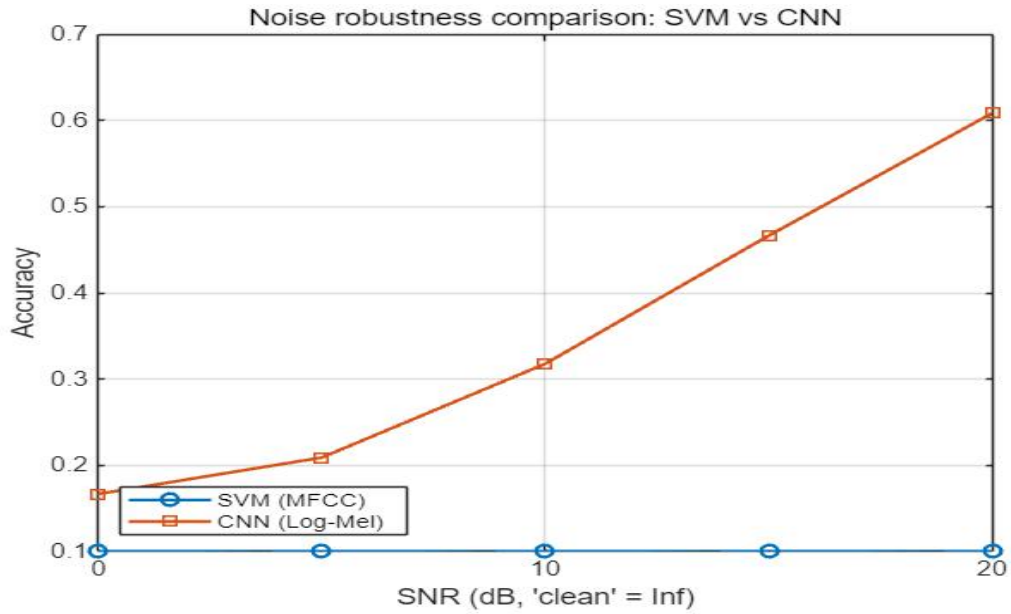


Fig4.8 Noise robustness comparison: SVM vs CNN

5. Discussion

5.1 Limitations of traditional MFCC+SVM

The poor performance of MFCC+SVM comes from its structural limitations in acoustic feature expression and temporal information processing. First, MFCC performs multiple compressions from the spectrum to the cepstral domain during the extraction process, greatly simplifying the original time-frequency structure, resulting in the weakening of many fine-grained formant shapes that are critical to distinguishing digital speech. Therefore, the acoustic differences between different digits are further narrowed in the feature space, making it difficult to separate the classification boundaries. Secondly, SVM can only process static vectors in nature and cannot model the time evolution characteristics of speech. The pronunciation mode of each FSDD digit has obvious temporal intensity changes and the rhythm differences of different speakers. These cannot be expressed by simple MFCC mean features, making SVM lack of structural advantages when facing naturally sequential signals such as speech. In addition, when the amount of data is small and noisy, the MFCC features will experience obvious distribution drift, further weakening the generalization ability of SVM, causing its accuracy in a noisy environment to almost drop to the level of random guessing (around 10%). In summary, the information loss of MFCC and the unsuitability of SVM for dynamic speech jointly limit the final performance of the model.

5.2 CNN has stronger learning ability for Log-Mel

In contrast, CNN combined with Log-Mel spectrum showed stronger feature learning and generalization capabilities in this study. The main reason is that the Log-Mel feature retains the two-dimensional time-frequency structure of the speech signal, allowing the model to directly observe the trajectory of the formants changing over time, thereby obtaining a better performance than MFCC. Richer acoustic information; convolutional neural networks have powerful local pattern capture capabilities and can identify key features such as energy mutations, formant trends, and transition structures within local time windows. These exactly correspond to the main acoustic cues for identifying numbers in human pronunciation. In addition, the translation invariance of the convolutional layer can effectively handle the pronunciation speed and slight pitch differences of different speakers, making the features more stable between speakers, thereby significantly enhancing the generalization ability of the model. In the noise experiment, the accuracy of CNN in the range of 0–20 dB showed an obvious monotonic increasing trend with the increase of SNR, indicating that it can separate robust local structural information from noise, rather than being completely overwhelmed by the overall noise energy perturbation like SVM. More importantly, the small network structure of CNN achieved an inference latency of less than 0.03 ms/sample in this project, proving that it not only has better performance but also has high real-time and engineering application potential.

5.3 Dataset size affects model performance

The Free Spoken Digit Dataset (FSDD) used in this study is small in size and contains only thousands of recordings from a few speakers. This "small sample, low diversity" data characteristic has a completely different impact on the two types of models. For SVM, it is difficult for small-scale data to cover the speech differences of all speakers, and the classification hyperplane learned by it is easy to overfit the training set, causing the model's performance to rapidly degrade in the test set and noisy environment; At the same time, the feature expression dimension of MFCC itself is relatively fixed and low-order, making it impossible to further mine differentiated structural information from limited data. For CNN, although the small amount of data will limit the model to achieve higher accuracy, the rich time-frequency structure provided by Log-Mel still enables it to learn effective local patterns from limited samples, allowing the model to maintain significantly better performance than SVM. However, the small sample data still causes the CNN to have certain confusion on similar phonemes such as the digits 0/9, 5/8, etc., and as the noise increases, the model performance is not as stable as the deep model in large-scale tasks. Therefore, the data size limits the generalization ability of this research model in more realistic and complex speech scenes, but it also highlights the structural advantages of CNN in small sample tasks.

5.4 Limitations

Although this study successfully compared two different speech recognition models, there are still several limitations that need to be further addressed in subsequent work. First, the FSDD data set is small in size and has a limited number of speakers, so the model may experience performance degradation when facing more complex acoustic changes in the real world, especially in zero-sample or cross-speaker scenarios. The generalization ability still needs to be verified. Secondly, the noise used in the study is only Additive White Gaussian Noise, which is different from the environmental noise in real speech (such as street noise, vocal background, reverberation, etc.). Therefore, the noise robustness conclusion of the model is mainly for the AWGN scene, and the applicability to real complex interference still needs additional testing. In addition, the CNN model used in this study has a relatively simple structure, focusing mainly on lightweight design and basic convolution capabilities. Therefore, more advanced methods such as attention mechanisms, deep networks, and data enhancement were not further explored, resulting in a large room for improvement in model performance. Furthermore, the impact of feature standardization and normalization methods was not studied in depth in the experiment, and these steps may significantly change the training stability and generalization ability of the model. Finally, since the project goal focuses on comparing two types of typical models, this study did not introduce end-to-end speech recognition technology (such as RNN, CRNN or Transformer). Therefore, the overall performance represents a baseline level of traditional feature engineering methods and basic deep learning architecture, rather than the optimal result in the current field of speech recognition. The above limitations all provide directions for further improvements in the future.

6. Conclusion

This study focuses on the small-sample speech recognition task and compares and analyzes the

performance of the traditional MFCC+SVM method and the CNN deep learning model based on Log-Mel spectrum under different noise conditions. The experimental results clearly show that in a clean speech environment, both methods can achieve high accuracy. However, when the noise gradually increases, the performance of SVM declines rapidly, while CNN can maintain higher accuracy more stably, showing obvious advantages in noise robustness. Through the analysis of the feature construction mechanism, the research further reveals that the strong compression characteristics of MFCC will lead to the loss of time-frequency structure, while Log-Mel can retain richer spectroscopic information, allowing the convolutional model to capture key signal patterns. In addition, through SNR-Accuracy curve and inference latency (Latency) measurements, this study verified that lightweight CNN is not only better than SVM in terms of accuracy, but also can meet the requirements of small speech recognition systems in terms of real-time performance.

Overall, this study completed a complete experimental chain from feature construction, model selection, noise robustness testing to real-time evaluation, providing systematic evidence for understanding the core differences between traditional methods and deep learning methods. Research results show that Log-Mel + CNN is an efficient speech recognition solution suitable for small data sets, while MFCC+SVM is more suitable as a baseline model. Ultimately, this research not only completed the goals of the classroom project, but also provided clear technical direction and practical reference for subsequent expansion to real-time speech recognition with multiple speakers, multiple categories, and more complex environments.

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